CROWN DENTAL & COSMETOLOGY CLINIC

Vibe Coding Master Design & Product Specification

Premium animated public website + responsive clinic management portal

Purpose: one source of truth for AI-assisted design and implementation. This follows the six-document method in the supplied Vibe Coding guide and uses the supplied resource library for motion, components, typography, color and production references.

1. PRD — Product Requirements Document

Field	Crown Dental Specification
App Name	Crown Dental Clinic Management & Patient Care Platform
Public tagline	Confident smiles. Thoughtful dental care. A modern patient experience.
Problem	Patients need a simple, trustworthy way to discover treatments, understand the clinic, request/book appointments and receive a consistent digital experience. Clinic staff need one secure system for patients, appointments, queues, schedules and clinical records.
Primary users	Public visitors/patients; Receptionists; Doctors; Administrators. Doctors can also submit a public application for admin approval.
Core value proposition	A dental-first platform connecting the public clinic experience with a focused internal workflow: Patient → Dental Chart → Treatment Plan, connected through Appointments and Consultations.
Must Have	Responsive public website; treatment pages; doctors; appointment booking/request flow; doctor application; authentication; receptionist patient/appointment/queue workflows; doctor patient history, dental chart, consultation, treatment plan, prescription and follow-up; admin management; RBAC; audit trail; notifications.
Nice to Have	Smile gallery with consent controls; before/after media; advanced dental chart interactions; analytics; enhanced motion scenes; AI-assisted FAQ/chat; richer reports.
Out of Scope for V1	Pharmacy, laboratory, radiology information system, inpatient/ward management, emergency/OT/blood-bank/ambulance modules, insurance claims, FHIR/ABDM integrations, advanced AI diagnosis.
Success Criteria	Core booking and clinic workflows work end-to-end; server-side permissions are enforced; primary screens are usable on mobile; animation never blocks clinical tasks; loading/empty/error states are designed; accessibility and performance checks pass.

2. TRD — Technical Requirements Document

Field	Crown Dental Specification
Frontend	React + Vite, responsive CSS, component-based architecture.
UI system	Tailwind CSS + shadcn/ui-style primitives; Lucide icons; reusable design tokens. Avoid raw/default browser controls.
Motion	Framer Motion / Motion for interface transitions and micro-

	interactions; GSAP only where timeline-level animation is genuinely needed; Lenis optional for controlled public-page smooth scrolling. Respect reduced-motion preferences.
Backend	Java + Spring Boot REST API; Spring Security; JWT authentication/authorization; Spring Data JPA/Hibernate.
Database	MySQL-compatible relational database.
Auth/session	JWT on backend. Frontend session state should use sessionStorage so separate browser tabs can hold independent sessions; backend authorization remains authoritative.
Storage	Secure object/file storage for patient documents, doctor applications, certificates, X-rays/images and approved public media. Private clinical files must not be public.
Notifications	Email notifications for appointment confirmations/reminders and relevant doctor/patient updates.
Deployment	Frontend and backend independently deployable; production secrets/configuration through environment variables.
Constraints	Mobile-first; keyboard accessible; fast initial load; animation degrades gracefully; clinical portal prioritizes clarity over decoration; no sensitive patient data in client logs.
Environment variables	API base URL; database URL/credentials; JWT secret; email provider credentials; storage credentials; allowed frontend origin; other provider keys only when required.

3. App Flow — Navigation & User Journey Map

3.1 Public navigation

- Home → Treatments → Treatment Detail → Book Appointment.
- Home → Doctors → Doctor Profile → Book Appointment.
- Home → Smile Gallery / Patient Stories → consent-controlled media.
- Home → About / Technology / FAQs / Contact.
- Doctor Application → Application Form → Submission Confirmation → Admin Review → Approved/Rejected.
- Login → role-aware redirect to Receptionist / Doctor / Admin dashboard.

3.2 Appointment journey

- Visitor selects treatment → optional doctor → preferred date/time → patient/contact details → review → confirmation.
- Receptionist can search/create patient → view doctor availability → create/reschedule/cancel appointment → check-in patient → add to queue.
- Doctor opens queue → selects patient → reviews history/dental chart → records consultation → creates/updates treatment plan → prescription/follow-up → marks consultation complete.

3.3 Doctor application journey

- Public applicant opens Doctor Application.
- Enter identity, contact, qualification, registration number, specialization, experience, department, availability and supporting documents.
- Submit → status PENDING.
- Admin reviews → approve or reject with audit record.
- Approval activates/creates doctor account and profile; rejection stores reason and status.

3.4 Role navigation

Field	Crown Dental Specification

Receptionist	Dashboard | Patients | Appointments | Doctor Availability | Queue | Documents | Profile
Doctor	Dashboard | Appointments | Patients | Patient 360 | Dental Chart | Consultation | Treatment Plans | Prescriptions | Documents | Follow-ups | Profile
Admin	Dashboard | Doctor Applications | Doctors | Receptionists | Patients | Departments | Treatments | Schedules | Appointments | Users & Roles | Reports | Audit Logs | Settings

3.5 Edge states

- Loading: skeletons matching the final layout; never a blank screen.
- Empty: concise explanation plus the primary next action.
- Error: human-readable message plus retry; preserve entered form data where safe.
- Unauthorized: redirect to login or show access-denied page.
- Session expiry: warn, preserve safe form state, re-authenticate.
- Mobile: tables become cards or controlled horizontal-scroll regions; sidebars collapse into drawer/bottom navigation.

4. UI/UX Design Brief — Visual & Interaction Design

Design direction: premium dental/cosmetology editorial website outside the portal, paired with a calm clinical SaaS interface inside. The supplied Vibe Coding guide says the design brief should lock visual language, colors, fonts, component style, spacing, interaction patterns, mobile behavior and accessibility before implementation.

4.1 Visual system

Field	Crown Dental Specification
Aesthetic	Premium, elegant, modern dental/cosmetology, editorial + clinical precision. Avoid generic AI-dashboard appearance.
Dominant background	Warm ivory/near-white with subtle depth; clean white surfaces for cards.
Primary brand direction	Deep charcoal/ink + refined dental teal/blue-green or muted royal accent.
Accent	One restrained premium accent for CTA, focus, active navigation and highlights.
Color ratio	Use the supplied resource library's 60–30–10 rule: 60% dominant background, 30% secondary/content surfaces, 10% brand accent.
Typography	Modern sans-serif such as Inter/Manrope for UI; tasteful display face only for large public headings.
Radius	Moderate 14–20px public cards/sections; 10–14px clinical controls. Avoid excessive pill UI.
Borders	Thin, low-contrast borders; elevation used sparingly.
Shadows	Soft ambient shadows only; no heavy drop shadows.
Density	Public pages: spacious/editorial. Internal portal: compact, information-dense but breathable.

4.2 Public website animation model

- Hero: staggered entrance for headline, copy, CTA and visual layer; no aggressive perpetual motion.
- Scroll reveal: small translate/opacity transitions, triggered once or carefully replayed.
- Treatment cards: image crop/scale and metadata transition without layout shift.
- Doctor cards: gentle image/overlay movement; CTA remains obvious.
- Appointment CTA: restrained hover/focus micro-interaction; no bouncing.
- Page transitions: short fade/slide between major routes.
- Gallery: smooth image transitions and lightbox; respect reduced-motion.
- Optional 3D/shader: lightweight decorative hero layer only; never required for navigation or booking.

- Smooth scroll: optional on desktop public pages; reduce/disable on mobile if it harms performance or native scrolling.
- Use Motion/GSAP/Lenis selectively; the resource library explicitly lists these motion references.

4.3 Component references

- Use shadcn/ui-style primitives for dialogs, dropdowns, tabs, forms, sheets and buttons.
- Use React Bits / Smooth UI / similar component references only when they fit the established design system; do not mix unrelated visual styles.
- Use Lucide icons consistently.
- The supplied resource library lists React Bits, Smooth UI, Kokonut UI, Flowbite and Preline as component references.

4.4 Mobile-first rules

- Design at 360px first, then validate 390, 430, tablet and desktop breakpoints.
- Public mobile header: logo + compact menu + primary booking CTA where space permits.
- Internal mobile navigation: compact top bar plus bottom navigation for highest-frequency actions; secondary actions in sheets/drawers.
- Appointment form: one-column, thumb-friendly controls, sticky bottom confirmation CTA only when it does not cover fields.
- Dental chart: horizontally navigable tooth map or focused tooth selector on small screens; never shrink labels until unreadable.
- Tables: convert to stacked cards or controlled horizontal scroll; never force desktop-width tables into tiny screens.
- Touch targets: minimum comfortable 44px interactive area.
- Respect safe areas and avoid fixed elements covering browser UI.

4.5 Accessibility & performance

- WCAG-oriented contrast and keyboard navigation.
- Visible focus states and semantic form labels/errors.
- prefers-reduced-motion: replace non-essential animation with minimal transitions.
- Lazy-load below-the-fold media; optimize images; use responsive image sizes.
- Animation is progressive enhancement, never the only way to understand information.
- The supplied resource library explicitly includes responsive design, performance and accessibility in optimization.

4.6 Public page design

- Home: animated hero → clinic introduction → treatment categories → doctor highlights → technology/experience → smile gallery → appointment CTA → location/contact → footer.
- Treatments: categories with compact cards and animated detail transitions.
- Treatment detail: hero, what it is, who it may be relevant to, process, expectations, FAQs, CTA.
- Doctors: elegant profile cards and detailed profiles using only clinic-verified credentials.
- Smile Gallery: consent-controlled before/after imagery with privacy handling.
- Book Appointment: short progressive form with clear progress indicator.
- Doctor Application: structured professional application with upload states and submission confirmation.

4.7 Internal portal design

- Calm clinical SaaS shell: sidebar on desktop, drawer/bottom navigation on mobile.
- Dashboard cards are functional: today's appointments, waiting, completed, follow-ups, pending applications where relevant.
- Patient 360 is the central clinical workspace.
- Dental Chart is a dedicated visual workspace tied to patient and treatment plan.
- Treatment Plan uses: Diagnosis → Planned → Scheduled → In Progress → Completed → Follow-up.
- Use drawers/modals for secondary edits so clinicians stay in context.
- Use compact motion for status changes, panel entrance, success confirmation and queue movement.

5. Backend Schema — Data Model & Auth

Field	Crown Dental Specification
User	id, name, email, phone, password credential reference, role, status, timestamps
Role	ADMIN, RECEPTIONIST, DOCTOR; permissions enforced server-side
Patient	id, patient_code, name, DOB, gender, phone, email, address, emergency_contact, medical_history_summary, dental_history_summary, timestamps
PatientDocument	id, patient_id, document_type, file_reference, uploaded_by, visibility, consent metadata, timestamps
Doctor	id, user_id, department_id, specialization, qualification, registration_number, experience, bio, photo, status
DoctorApplication	id, applicant details, specialization, qualification, registration_number, experience, department, availability, documents, status, review_reason, reviewed_by, timestamps
Department	id, name, description, status
Treatment	id, name, category, description, public_content, status
DoctorSchedule	id, doctor_id, day, start_time, end_time, slot_duration, status
Appointment	id, patient_id, doctor_id, treatment_id, appointment_date, slot, status, source, notes, created_by, timestamps
Consultation/MedicalRecord	id, appointment_id, patient_id, doctor_id, symptoms, examination, diagnosis, clinical_notes, attachments, timestamps
DentalChartEntry	id, patient_id, tooth_number, condition, finding, procedure, notes, updated_by, timestamps
TreatmentPlan	id, patient_id, doctor_id, diagnosis, status, created_at, updated_at
TreatmentPlanItem	id, treatment_plan_id, tooth_number, treatment, status, scheduled_appointment_id, notes
Prescription	id, consultation_id, doctor_id, patient_id, instructions, timestamps
PrescriptionItem	id, prescription_id, medicine_name, dosage, frequency, duration, instructions
Notification	id, user/patient reference, type, channel, message, status, scheduled_at, sent_at
AuditLog	id, actor_user_id, action, entity_type, entity_id, timestamp, metadata

5.1 Security rules

- RBAC is mandatory on backend endpoints and service methods; frontend route guards are not a security boundary.
- Receptionist: patient administrative data, appointments, schedules, queue and permitted documents; no editing of doctor clinical notes/dental findings.
- Doctor: assigned clinical workflow, patient history, dental chart, consultation, treatment plan, prescriptions and follow-ups; no user/system administration.
- Admin: operational configuration and user/application management; sensitive clinical access remains audited.
- All clinical changes record actor and timestamp.
- Private documents are access-controlled; public media requires explicit publication/consent workflow.

6. Implementation Plan — Step-by-Step Build Sequence

Field	Crown Dental Specification
Phase 1 — Foundation	React/Vite frontend, Spring Boot backend, MySQL, environment variables, folder conventions, shared design

	tokens, linting/formatting.
Phase 2 — Data & Security	Entities/migrations, authentication, JWT, RBAC, validation, audit model, secure file access.
Phase 3 — Design System	Tokens, typography, buttons, forms, cards, tables, dialogs, sheets, navigation shell, skeletons, error/empty states, responsive primitives.
Phase 4 — Public Website	Home, treatments, treatment detail, doctors, doctor profile, gallery, FAQs, contact, appointment flow, doctor application.
Phase 5 — Receptionist	Dashboard, patient registration/search, appointment management, doctor availability, queue/check-in, documents.
Phase 6 — Doctor	Dashboard, appointments, Patient 360, dental chart, consultation, treatment plans, prescriptions, documents, follow-ups.
Phase 7 — Admin	Dashboard, applications, doctors, receptionists, patients, departments, treatments, schedules, appointments, users/roles, reports, audit logs, settings.
Phase 8 — Motion & Polish	Public animation choreography, micro-interactions, route transitions, dental-chart interactions and responsive refinements. Motion is progressive enhancement.
Phase 9 — Testing & QA	Role flows, permissions, mobile layouts, keyboard navigation, reduced motion, loading/error/empty states, appointment conflicts, document access and audit trails.
Phase 10 — Deployment	Production environment, domain/origin configuration, secrets, database backup strategy, email/storage configuration, production build, SEO/social metadata and final QA.

6.1 Definition of done

- Every core user journey works from entry to completion.
- Every role sees only permitted actions.
- Every major public page has intentional responsive behavior.
- Every major portal screen works at mobile width without broken overflow.
- Animation is smooth, purposeful and reduced when accessibility settings request it.
- No placeholder lorem ipsum or random generated UI remains.
- Clinical data and files are protected and auditable.
- Production build passes functional, responsive and accessibility checks.

7. AI Build Instruction — Source of Truth

Use this document as the source of truth for the Crown Dental & Cosmetology Clinic platform. Do not invent screens, roles, permissions, clinical workflows, colors, animations or dependencies when a decision is already defined here. If an implementation detail is genuinely missing, identify the gap before making a structural decision.

- Build the public website and internal clinic portal as one coherent product, while keeping their visual density distinct.
- Prioritize mobile usability before desktop polish.
- Use animation to communicate hierarchy, continuity and state—not visual noise.
- Do not introduce a generic hospital-dashboard aesthetic.
- Keep dental-specific objects central: Patient, Dental Chart, Treatment Plan, Appointment and Consultation.
- Use reusable components and design tokens; avoid page-specific CSS hacks.
- Before adding 3D or heavy effects, verify bundle/performance impact and whether CSS/Motion is sufficient.

8. Resource Reference Mapping

The supplied resource library recommends a research → design → assets → build → components → motion → media → marketing → SEO → optimization → launch workflow.

Field	Crown Dental Specification
Research / inspiration	Mobbin, Godly, Awwwards, Dribbble, Behance, DesignVault
Design / prototyping	Figma, Stitch, Motiff, Visily, Uizard, Galileo
Motion	Motion.dev, GSAP, Lenis, Anime.js, Vanta, Motionsites
Components	React Bits, Smooth UI, Kokonut UI, Flowbite, Preline
Visual assets	Haikei, ThreeUI, Pacdora, Shaders, Neurascapes
Optimization	Sharp.js, responsive design, performance, accessibility
Launch / SEO	Search Console, Bing Webmaster Tools, sitemap, IndexNow, Open Graph

END OF MASTER SPECIFICATION